

ARAP · AIRNAV RISK ANALYSIS PLATFORM

User Guide

How to plan, run, review and approve safety risk assessments
with the twelve methods of the AirNav Risk Analysis Manual

Software version 0.2 · Guide edition 2 · September 2026

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ARAP User Guide

AirNav Risk Analysis Platform — version 0.2

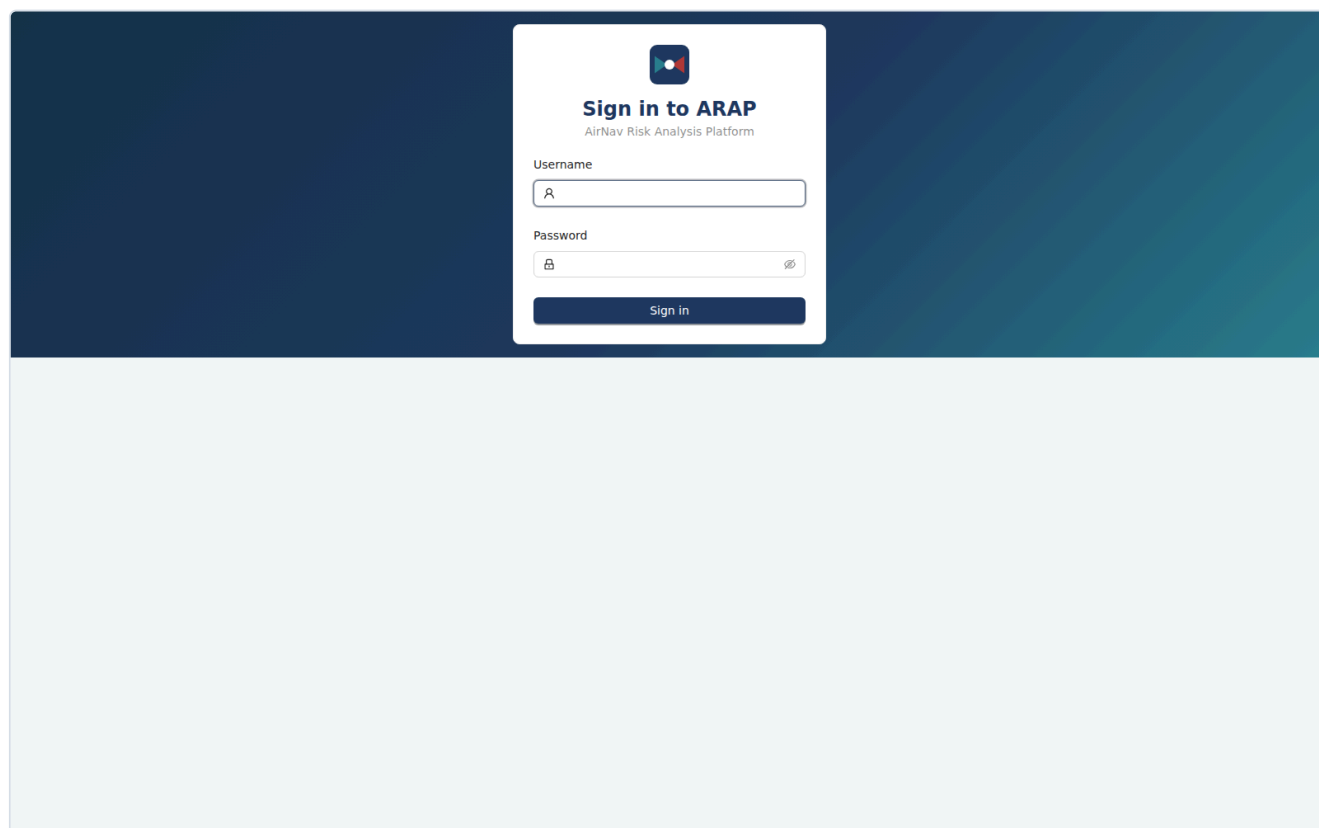
ARAP is the web application that implements the *AirNav Risk Analysis Manual*. It lets AirNav staff plan, run, review and approve safety risk assessments with twenty-five methods, keeps every hazard in one hazard log, and produces the Safety Assessment Report.

This guide explains how to use the application. For *why* and *when* to use each method, see the Manual ([docs/AirNav_Risk_Analysis_Manual.pdf](#)). For the formal requirements, see the SRS ([docs/AirNav_Risk_Analysis_Software_Requirements.pdf](#)).

1. Getting started

1.1 Signing in

Open ARAP in a browser (Chrome, Edge, Firefox or Safari) at the address your administrator gives you — for a local installation this is `http://localhost:8080` . Enter your username and password.



Sign-in page

1.2 Roles

What you can do depends on your role:

Role	Can do
Viewer	Read everything in their scope; update actions they own.
Assessor	Create projects, assessments and method studies; edit worksheets and diagrams; send hazards to the hazard log; raise actions; submit assessments for review.
Reviewer	Everything an assessor can, plus return, endorse or reject assessments, and read the audit trail.
Authority	Accept (or reject) residual risk, but only for the risk regions assigned to them (for example <i>Tolerable (lower)</i>).
Admin	Manage users, roles and the risk classification scheme. Admin is not a substitute for the proper acceptance authority in normal use.

1.3 Demo accounts

When ARAP is installed with demo data (`ARAP_SEED_DEMO=true`), these accounts exist, all with password `demo1234` :

Username	Role	May accept
assessor	Assessor	—
reviewer	Reviewer	—
director	Authority	Tolerable (lower), Acceptable
accexec	Authority (Accountable Executive)	Tolerable (upper), Tolerable (lower), Acceptable
viewer	Viewer	—

The administrator account is `admin` ; its password is set at installation. **Disable the demo data and change all passwords before real use.**

1.4 Screen layout and language

- The **left menu** gives access to the Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme and User guide (plus Users and Audit trail for administrators and reviewers).
- The **EN / ID** switch at the top right changes the interface language between English and Bahasa Indonesia. Your choice is remembered on that computer. Text you type into worksheets is stored as entered.
- Your name and role are shown top right; click them to sign out.

3. Projects and assessments

3.1 Projects

A **project** represents one change: a new system, a procedure, an airspace change, an organisational change or a temporary change. Go to **Projects & assessments** → **New project** and enter a code (e.g. ADSB-2027), a title, the type of change, the units affected and the sponsor.

The screenshot shows the ARAP (AirNav Risk Analysis Platform) interface. The sidebar on the left contains navigation links: Dashboard, Projects & assessments (selected), Hazard log, Actions, Risk scheme, and User guide. The main content area is titled 'Projects & assessments' and includes a '+ New project' button. Below the title, there is a description: 'A project is a change (system, procedure, airspace, organisational). Each has one or more safety assessments.' The 'Projects' section displays a table with the following data:

Code	Title	Type	Units	Assessments
APP-RELOC	Relocation of Jakarta approach unit to new ATC centre	organisational	Jakarta APP	1
DEMO-01	Manual worked examples	system	Jakarta ACC; Soekarno-Hatta TWR	1

The 'All assessments' section displays a table with the following data:

Project	Assessment	Version	Status	Updated
APP-RELOC	Transition safety assessment	1	in review	2026-09-28 10:22
DEMO-01	Worked examples — all twelve methods	1	draft	2026-09-28 10:22

At the bottom right of the 'All assessments' table, there is a pagination control showing '< 1 >'.

Projects

3.2 Assessments

Open the project and choose **New assessment**. Describe the **scope and system boundary**, the **operational environment** and your **assumptions** (Manual §2.3). The assessment records which version of the risk scheme it uses.

The assessment page brings everything together:

Worked examples — all twelve methods draft v1

Submit for review | Report (.docx)

Scope, environment and assumptions Edit

Scope: Demonstration of every ARAP method module using the Manual's examples.
 Environment: Illustrative values only.
 Assumptions: All numbers are illustrative (Manual, How to use).
 Risk scheme: version 1

Method studies (12) Add study

Suggested for a system change (Manual §4.1): [HAZID](#) [FHA](#) [FMEA / FMECA](#) [FTA](#) [Bowtie analysis](#)

Method	Study	Status	Updated
Bowtie analysis	Runway incursion bowtie (Manual §6)	open	2026-09-28 10:22
HAZID	HAZID — approach unit relocation (Manual §7.4)	open	2026-09-28 10:22
HAZOP	HAZOP — estimate message coordination (Manual §8.3)	open	2026-09-28 10:22
JHA	JHA — obstruction light on radar tower (Manual §9.3)	open	2026-09-28 10:22
FMEA / FMECA	FMEA — sector VHF air-ground (Manual §10.3)	open	2026-09-28 10:22
LOPA	LOPA — level bust (Manual §11.3)	open	2026-09-28 10:22
FHA	FHA — ADS-B en-route surveillance (Manual §12.3)	open	2026-09-28 10:22
STPA	STPA — descent clearances (Manual §13.2)	open	2026-09-28 10:22
FTA	FTA — loss of surveillance picture (Manual §14.3)	open	2026-09-28 10:22
Fatigue modeling	Fatigue — night shift with/without nap (Manual §15.4)	open	2026-09-28 10:22
FRAM	FRAM — descent clearances in a busy arrival sector (Manual §16)	open	2026-09-28 10:22
Bayesian belief network	BBN — coordination error and loss of separation (Manual §17.3)	open	2026-09-28 10:22

Hazards (5) Open hazard log →

Ref	Hazard	Initial	Residual	Controls
HZ-0001	Unauthorised entry onto active runway	3A	2A	2
HZ-0002	Level bust leading to potential mid-air collision	3A	3A	2
HZ-0003	Total loss of surveillance picture at ACC sector	3B	3B	1
HZ-0004	ATCO fatigue at end of night shift	4C	3C	1
HZ-0005	New site outside radio line of sight of low-level sector	3B	—	0

Risk acceptance

Worst current/residual region: [Tolerable \(upper\)](#)
 Acceptance authority: **Accountable Executive**

Actions (5) All actions →

- ACT-0001 Install diverse, physically separate surveillance data path
CNS engineering - due 2026-12-27 - [open](#)
- ACT-0002 Implement SFL mismatch alert in ATM system
ATIS systems - due 2027-02-25 - [open](#)
- ACT-0003 Collect KSS survey for two roster cycles
FRMS specialist - due 2026-09-18 - [open](#)
- ACT-0004 Audit stop-bar U/S procedure compliance
Aerodrome ops - due 2026-09-25 - [in progress](#)
- ACT-0005 Radio coverage study for new site
CNS engineering - due 2026-10-28 - [open](#)

Review and approval history

Not yet submitted.

Assessment page

- **Method studies** — the analyses carried out for this assessment. ARAP suggests methods for the type of change (Manual §4.1); the suggestion is only advice.
- **Hazards** — hazard-log entries that belong to this assessment, with initial and residual risk.
- **Risk acceptance** — the worst residual risk region and the authority that must accept it.
- **Actions** and the **review and approval history**.
- Workflow buttons at the top right (see [section 15](#)) and **Report (.docx)**.

3.3 Adding a method study

On the assessment page choose **Add study**, pick the method (suggested methods are marked ★) and give the study a title, e.g. *HAZOP of cut-over plan*. The study opens straight away.

4. Working in a method study

Every study page has the same header:

- The **method** tag and the **study title** (editable).
- A link back to the assessment, the Manual chapter for the method and the template version.
- The number of **participants** recorded.
- **Save**. Changes are not saved automatically: the button reads *Save* while there are unsaved changes and *Saved* afterwards. If you try to leave the page with unsaved changes ARAP asks you to confirm.

When the assessment is endorsed, accepted or closed, or if your role cannot edit, the study opens **read-only** and a blue banner says so.

4.1 Sending results to the hazard log

Every method has a **Send to hazard log** button. It creates a hazard-log entry for each selected worksheet row (or for the bowtie as a whole, or for each STPA hazard), carrying over the causes, consequences, controls and any severity and likelihood. Sending the same row again **updates** the existing hazard rather than creating a duplicate. A message lists the hazard references created, e.g. HZ-0007, HZ-0008 . Rate the risk and complete the controls in the [hazard log](#).

5. Worksheet methods

HAZID, HAZOP, JHA, FMEA/FMECA and FHA use a spreadsheet-style worksheet.

Editing the worksheet

- **Double-click** a cell (or press Enter) to edit it; press Enter or click elsewhere to finish.
- Columns with fixed values — severity, likelihood, guideword, control type, failure type — show a drop-down list.
- **Add row** appends a row with the next ID (HZ-06 , HP-04 , ...).
- Tick the checkboxes on the left to select rows, then **Delete selected** or **Send ... to hazard log**.
- Click a column header to sort; use the filter icon to filter.
- The **Risk** column is computed from severity and likelihood using the current risk matrix.

ARAP AirNav Risk Analysis Platform

EN ID Ayu Assessor assessor

HAZID HAZID — approach unit relocation (Manual §7.4)

Worked examples — all twelve methods · Manual chapter 7 · template v1.0

2 participants Saved

+ Add row Delete selected Send 2 selected to hazard log Double-click a cell to edit. Tick rows to select.

ID	Guideword	Hazard	Cause(s)	Consequence	Existing controls	S	L	Risk	Actions
HZ-01	Transition	Loss of surveillance display at cut-over	Data feed not re-routed; configuration error	Loss of separation during cut-over	Cut-over at low traffic; old site on hot standby	C	3	3C	HAZOP of cut rollback criteria
HZ-02	People	ATCOs unfamiliar with new CWP HMI	Short training window	Delayed or erroneous inputs	Simulator training	D	4	4D	Minimum hours before live ops
HZ-03	Interfaces	Wrong telephone coordination lines to TWR	VCCS mapping error	Late coordination of missed approach	Test plan	C	3	3C	End-to-end line each TWR
HZ-04	Equipment / systems	Single power feed to new building	Design	Total loss of service	UPS 30 min	B	2	2B	Second feed of power
HZ-05	Environment	New site outside radio line of sight of low-level sector	Topography	Loss of VHF contact below FL050	None yet	B	3	3B	Coverage study site

Guideword coverage

People: 1 Procedures: 0 Equipment / systems: 1 Airspace and aerodrome: 0 Traffic: 0 Environment: 1 Interfaces: 1 Transition: 1 Organisation: 0

Record "no hazard identified" for a guideword

HAZID worksheet after sending two rows to the hazard log

5.1 HAZID (Manual §7)

Work through the guidewords (People, Procedures, Equipment/systems, Airspace and aerodrome, Traffic, Environment, Interfaces, Transition, Organisation). The **Guideword coverage** panel below the worksheet shows how many rows each guideword has. When a guideword produces nothing, record that explicitly with **Record “no hazard identified”**. Use the *Needs method* column to flag rows that need a detailed method.

5.2 HAZOP (Manual §8)

1. In **Nodes and design intent**, add a node per procedure step or message exchange (+ **Node**), choose its type (*procedural* or *data-flow*) and write the design intent.
2. Press **Deviations** on a node. Choose the parameters and guidewords; ARAP adds one worksheet row per combination (e.g. *Message — No / Not*).

- For each row either tick **N/A** (not meaningful) or record causes, consequences, safeguards, S, L and a recommendation.
- The **Progress** column counts rows analysed out of rows generated.

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HAZOP HAZOP — estimate message coordination (Manual §8.3)

← Worked examples — all twelve methods · Manual chapter 8 · template v1.0

2 participants Saved

Nodes and design intent + Node

ID	Node	Type	Design intent	Progress
N3	Node 3: EST/CPL message upstream → downstream	data-flow	Downstream sector receives complete, correct and timely data for every flight crossing the boundary.	3/3 analysed Deviations

+ Add row Delete selected Send all rows to hazard log Double-click a cell to edit. Tick rows to select.

ID	Node	Parameter	Guideword	Deviation	Possible causes	Consequences	Safeguards	S	L
HP-01	N3	Message	No / Not	Estimate not received	Link failure; flight plan not activated; address error	Unknown traffic enters sector	Link-status alarm; verbal coordination backup	C	3
HP-02	N3	Content	Other than	Coordinated level differs from actual	Level changed after EST; no revision	Level conflict downstream	Mode C/S check; STCA	B	3
HP-03	N3	Timing	Late	Estimate inside agreed time	Short-notice route change	Workload	LoA fallback; verbal	D	4

HAZOP worksheet with node panel

5.3 JHA (Manual §9)

Fill in the job header (job, location, permits), then one row per job step with its hazards, controls, control type and the person responsible. ARAP warns when a step relies only on **Administrative** controls or **PPE**, the lowest levels of the hierarchy of controls.

5.4 FMEA / FMECA (Manual §10)

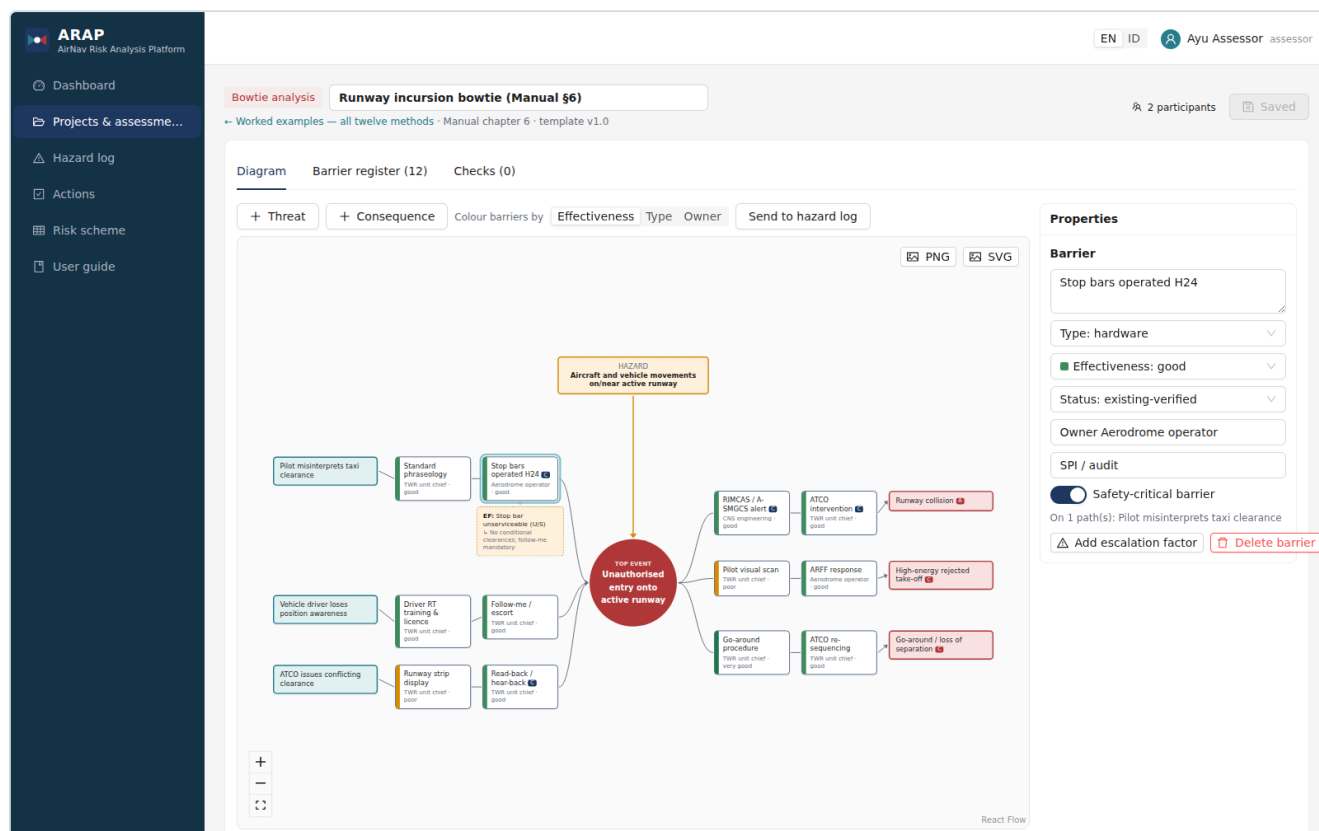
One row per failure mode: item, failure mode, cause, end effect on the ATS, detection, compensation, and **S, O, D** ratings (1–10). ARAP computes the **RPN** ($S \times O \times D$), colours high values and sorts by RPN. It also lists high-severity modes ($S \geq 8$) so they are not missed because of a low RPN. For FMECA, add the failure rate λ_p , mode ratio α , conditional probability β and operating time t .

5.5 FHA (Manual §12)

One row per failure condition: the function, the failure type (total loss, partial loss, erroneous detected/undetected, unintended, delayed), the condition, the operational effect and the **severity**. The **Safety objective** column is computed from the risk scheme: the maximum frequency per operating hour that keeps the failure condition at or better than *Tolerable (lower)* (Manual Table 12.1). Record the derived safety requirements in the last column.

6. Bowtie analysis

The bowtie editor draws the diagram automatically: threats on the left, the top event in the centre, consequences on the right, barriers in order on each line and escalation factors below the barrier they weaken.



Bowtie editor with a barrier selected

Building the bowtie

1. Click the **HAZARD** box and the **TOP EVENT** circle to edit their text. Phrase the top event as the moment control is lost, *before* any damage.
2. **+ Threat** and **+ Consequence** add lines. Click one to edit its text (and a consequence's severity).
3. With a threat or consequence selected, the **Properties** panel lists its barriers in order: - **New barrier on this path** adds a barrier; - **Add an existing (shared) barrier** puts a barrier that already exists on this line too — a shared barrier shows **xn**; - the arrows reorder barriers and the bin removes a barrier from this line only.
4. Click a **barrier** to set its type, effectiveness, status (existing-verified, existing-unverified, planned), owner, SPI/audit reference and whether it is **safety-critical** (C badge).
5. With a barrier selected, **Add escalation factor** attaches a condition that degrades it. Select the escalation factor to write its escalation-factor barriers, one per line.

Viewing

- **Colour barriers by** effectiveness (default), type or owner.
- Zoom with the mouse wheel or the **+** / **-** controls; drag the background to pan; the frame icon fits the diagram.
- **PNG** and **SVG** download the full diagram as an image.
- **Barrier register** lists every barrier with its attributes (Manual §6.3).

- **Checks** lists issues: a top event phrased like a consequence, lines with fewer than two barriers, barriers without an owner.

ARAP

AirNav Risk Analysis Platform

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Bowtie analysis

Runway incursion bowtie (Manual §6)

← Worked examples — all twelve methods · Manual chapter 6 · template v1.0

2 participants

Saved

Diagram

Barrier register (12)

Checks (0)

Barrier	Side	Type	Owner	Effectiveness	Critical	Status	Paths	SPI / audit
Standard phraseology	prevention	human	TWR unit chief	good		existing-verified	Pilot misinterprets taxi clearance	
Stop bars operated H24	prevention	hardware	Aerodrome operator	good	Yes	existing-verified	Pilot misinterprets taxi clearance	
Driver RT training & licence	prevention	human	TWR unit chief	good		existing-verified	Vehicle driver loses position awareness	
Follow-me / escort	prevention	human	TWR unit chief	good		existing-verified	Vehicle driver loses position awareness	
Runway strip display	prevention	human-hardware	TWR unit chief	poor		existing-verified	ATCO issues conflicting clearance	
Read-back / hear-back	prevention	human	TWR unit chief	good	Yes	existing-verified	ATCO issues conflicting clearance	
RIMCAS / A-SMGCS alert	recovery	hardware	CNS engineering	good	Yes	existing-verified	Runway collision	
ATCO intervention	recovery	human	TWR unit chief	good	Yes	existing-verified	Runway collision	
Pilot visual scan	recovery	human	TWR unit chief	poor		existing-verified	High-energy rejected take-off	
ARFF response	recovery	human-hardware	Aerodrome operator	good		existing-verified	High-energy rejected take-off	
Go-around procedure	recovery	human	TWR unit chief	very good		existing-verified	Go-around / loss of separation	
ATCO re-sequencing	recovery	human	TWR unit chief	good		existing-verified	Go-around / loss of separation	

Barrier register

7. LOPA

Risk scheme

User guide

Scenario Level bust leading to a potential mid-air collision

Initiating event Level bust in sector X

f(IE) 0.5

Unit per year

Data source Illustrative

Tolerable target 0.00001

Enabling conditions / conditional modifiers

+ Add

Condition	Probability
Conflicting traffic present	0.05

+ Add

Safeguards — credited only if all four IPL criteria are met

+ Add

Safeguard	PFD	indep.	effec.	depen.	audit.	Shared dependencies	Justi
ATCO monitoring and intervention	0.10	☒	☒	☒	☒	surveillance	
STCA alert and ATCO resolution	0.10	☒	☒	☒	☒	surveillance	
ACAS II RA followed	0.10	☒	☒	☒	☒	e.g. surveillance	
SFL mismatch alert (proposed)	0.10	☒	☒	☐	☒	e.g. surveillance	
Recurrent RT training	0.10	☐	☐	☒	☒	e.g. surveillance	

Calculate

Mitigated frequency

2.50e-5 per year

Target

1.00e-5 per year

Verdict

Gap x2.5

Further reduction needed

x2.5 (0.40 orders of magnitude)

Credited IPLs

ATCO monitoring and intervention; STCA alert and ATCO resolution; ACAS II RA followed

Not credited: SFL mismatch alert (proposed)

Missing: dependable

Not credited: Recurrent RT training

Missing: independent, effective

IPLs ATCO monitoring and intervention, STCA alert and ATCO resolution share dependency 'surveillance' — independence must be justified

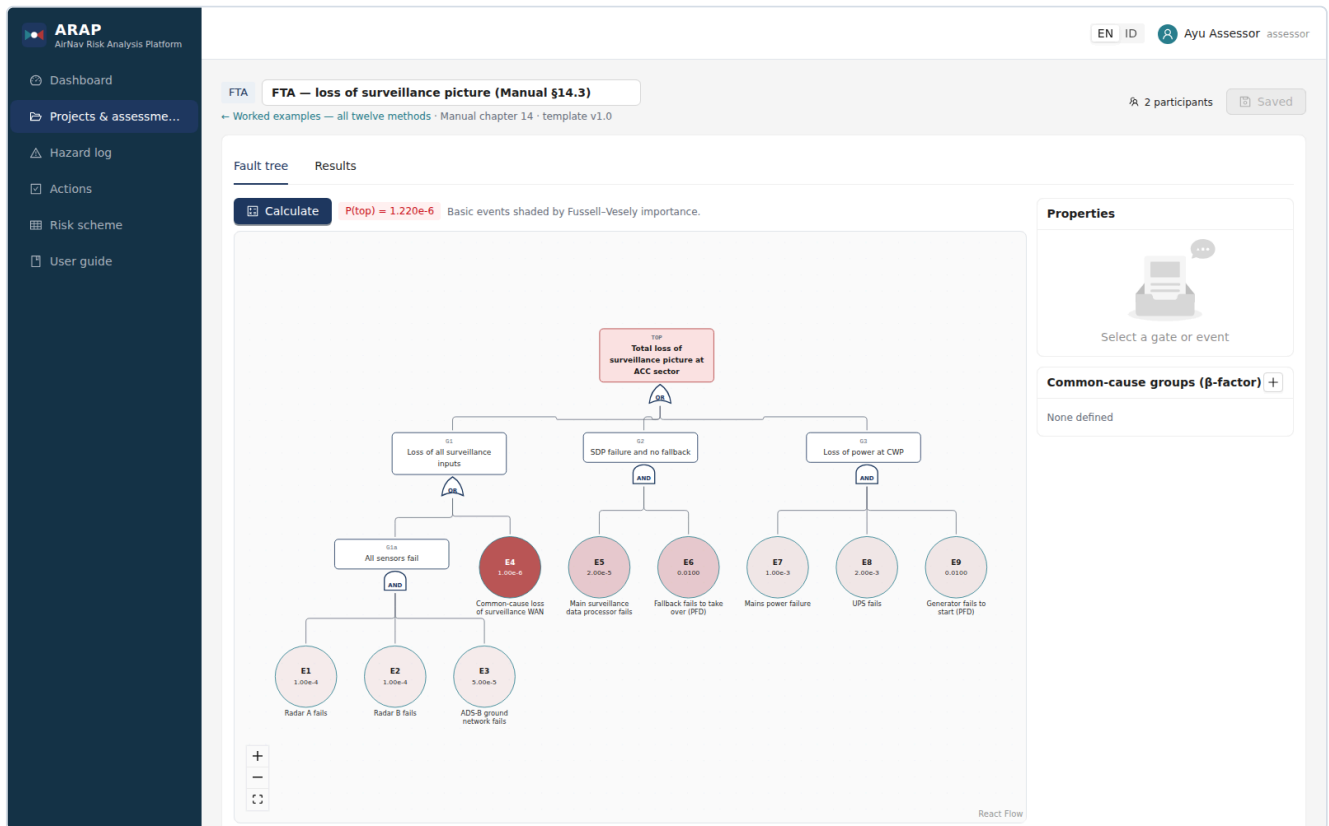
Layer	Frequency (per year)
Initiating event	5.0e-1
Conflicting traffic present	2.5e-2
ATCO monitoring and intervention	2.5e-3
STCA alert and ATCO resolution	2.5e-4
ACAS II RA followed	2.5e-5

LOPA calculator

- Describe the **scenario**, the **initiating event**, its frequency, unit and data source, and the **tolerable target** frequency.
- Add **enabling conditions / conditional modifiers** with their probabilities.
- Add **safeguards** with a probability of failure on demand (PFD). Tick each of the four IPL criteria — *independent, effective, dependable, auditable* — only if it is justified, and write the justification. A safeguard is credited **only if all four are ticked**.
- Record shared dependencies (e.g. `surveillance`) — ARAP warns if two credited layers share one.
- Press **Calculate**. The result shows the mitigated frequency, the verdict (*Target met* or *Gap x n*), the extra risk reduction needed, the safeguards not credited and why, and a log-scale chart of the frequency after each layer.

In the demo study the proposed *SFL mismatch alert* is not yet ticked *dependable*, so the result shows a gap of x2.5. Tick it and recalculate to see the target met (2.5×10^{-6} per year).

8. Fault tree analysis



Fault tree

Building the tree

- Click a gate and use **OR gate**, **AND gate**, **Basic event** or **Undeveloped** in the Properties panel to add inputs. **Reuse an existing node as input** links a node already in the tree (e.g. a shared support system).
- For a **vote** gate set k (k -out-of- n).
- For a **basic event** give either a probability p , or a failure rate λ with an exposure time t ($P = 1 - e^{-\lambda t}$), a repair time MTTR (repairable unavailability), or a test interval T (periodically tested standby, $\lambda T/2$).
- A **house event** is a switch set to TRUE or FALSE.
- Common-cause groups** apply the β -factor model to a set of basic events.

Calculating

Press **Calculate**. The top-event probability appears above the tree, and basic events are shaded by their Fussell-Vesely importance. The **Results** tab gives:

- the exact top-event probability (binary decision diagram), the rare-event approximation and the min-cut upper bound;
- single points of failure;
- minimal cut sets with probability and share;
- Birnbaum and Fussell-Vesely importance for every basic event.

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FTA — loss of surveillance picture (Manual §14.3)

Worked examples — all twelve methods · Manual chapter 14 · template v1.0

2 participants

Save

Fault treeResults

Top-event probability (exact, BDD)	1.2200e-6
Rare-event approximation	1.2200e-6
Min-cut upper bound	1.2200e-6
Single points of failure	E4
Engine	fta-1.1.0 · 31 BDD nodes

Minimal cut sets

Cut set	Order	Probability	Share
{E4}	1	1.00e-6	82.0%
{E5, E6}	2	2.00e-7	16.4%
{E7, E8, E9}	3	2.00e-8	1.6%
{E1, E2, E3}	3	5.00e-13	0.0%

Importance measures

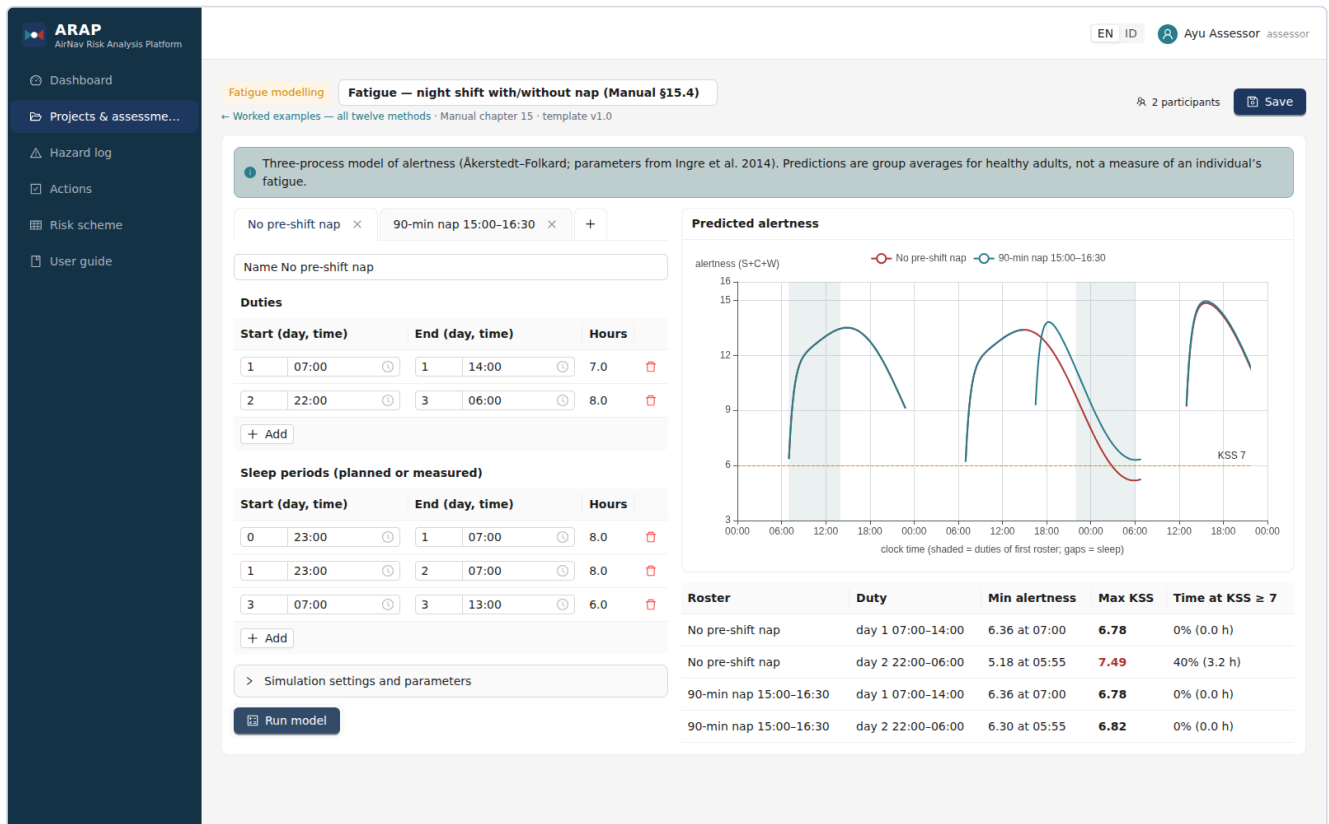
Event	Description	p	Birnbaum	Fussell-Vesely
E4	Common-cause loss of surveillance WAN	1.00e-6	1.00	0.820
E5	Main surveillance data processor fails	2.00e-5	1.00e-2	0.164
E6	Fallback fails to take over (PFD)	0.0100	2.00e-5	0.164
E7	Mains power failure	1.00e-3	2.00e-5	0.0164
E8	UPS fails	2.00e-3	1.00e-5	0.0164
E9	Generator fails to start (PFD)	0.0100	2.00e-6	0.0164
E1	Radar A fails	1.00e-4	5.00e-9	4.098e-7
E2	Radar B fails	1.00e-4	5.00e-9	4.098e-7
E3	ADS-B ground network fails	5.00e-5	1.00e-8	4.098e-7

FTA results

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9. Fatigue modelling



Fatigue model

1. Each tab is a **roster variant** (e.g. *No pre-shift nap* and *90-min nap*). Use **+** to copy the first roster into a new variant.
2. Enter **duties** and **sleep periods** as day number and clock time. Day 1 starts at 00:00; a sleep starting at 23:00 the evening before day 1 is entered as day 0, 23:00.
3. **Simulation settings and parameters** sets the number of days, time step, KSS threshold and, if needed, the model parameters (leave blank for the defaults of the Manual, §15.2).
4. Press **Run model**. The chart shows predicted alertness for each variant (gaps are sleep; shaded bands are duties of the first variant; the dashed line is the KSS threshold). The table gives, per duty, the minimum alertness and when it occurs, the maximum KSS, and the time spent at or above the threshold.

Predictions are group averages for healthy adults. They are not a measure of an individual's fatigue and must be checked against operational data (Manual §15.5).

10. Bayesian belief networks

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Bayesian belief network **BBN — coordination error and loss of separation (Manu)** 2 participants Save

← Worked examples — all twelve methods · Manual chapter 17 · template v1.0

Network

+ Node Update beliefs

Drag from a node's bottom handle to another node to add a causal arc. Set evidence in the panel.

ATCO fatigue F
yes 39.01%
no 60.99%

Workload W
high 54.48%
normal 45.52%

Coordination error E
yes 100.00%
no 0.00%

STCA available S
yes 89.89%
no 10.91%

Loss of separation L
yes 100.00%
no 0.00%

Node L

Label Loss of separation

States yes, no

Evidence yes

Conditional probability table

Coordination error	STCA available	P(yes)	P(no)	Σ
yes	yes	0.050	0.950	1.0
yes	no	0.300	0.700	1.0
no	yes	0.000	1.000	1.0
no	no	0.000	1.000	1.0

Sensitivity of P(yes) Delete node

P(evidence) = 2.453e-4 · engine bbn-1.0.0

Bayesian network with evidence

Building the network

- **+ Node** adds a node; drag nodes to arrange them.
- To add a causal arc, drag from the **bottom handle** of the cause to the effect node. Double-click an arc to remove it. ARAP rejects cycles when calculating.
- Select a node to edit its label and **states** (comma-separated) and its **conditional probability table** — one row per combination of parent states. Each row must sum to 1; the Σ column turns red when it does not (for two-state nodes the second value fills in automatically).

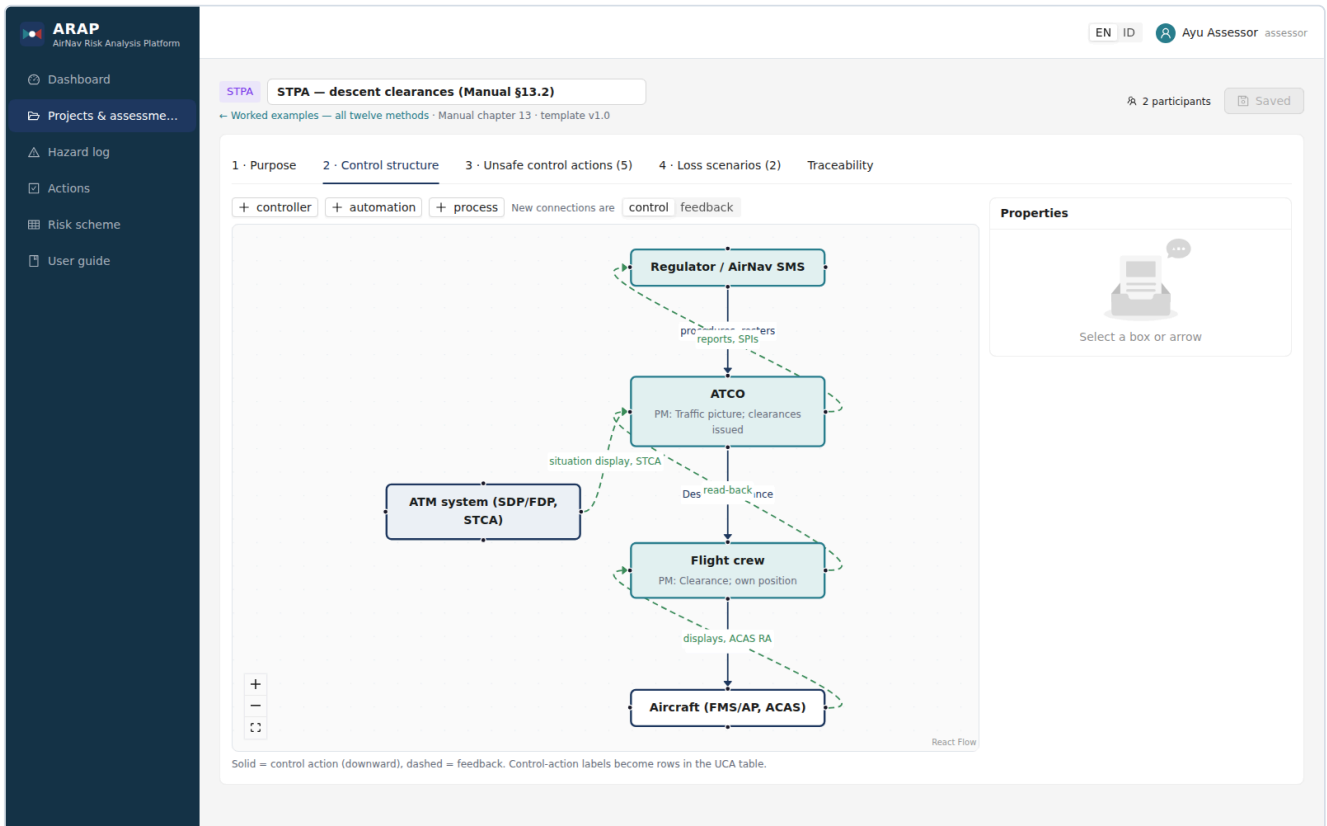
Using the network

- **Update beliefs** calculates every node's probabilities; the bars inside the nodes show them.
- Select a node and choose a state under **Evidence** to condition on it (the node turns amber). All other nodes update — this is how you ask diagnostic questions such as *given a loss of separation, how likely was fatigue?*
- **Sensitivity** shows how much each other node can move the selected node's probability (tornado chart).

11. STPA

The STPA study follows the four steps of the STPA Handbook, one tab each.

1. **Purpose** — losses, system-level hazards (linked to losses) and system-level constraints (linked to hazards).
2. **Control structure** — add controllers, automation and controlled processes; drag from a box's bottom/right handle to another box to add an arrow. Choose whether new arrows are **control** actions (solid) or **feedback** (dashed). Click a box to write its *process model*; click an arrow to rename it.



STPA control structure

3. **Unsafe control actions** — for every control action, four columns (*not providing, providing, too early/too late/wrong order, stopped too soon/applied too long*). Add UCAs with + **UCA**; each needs a context ("when ...") and at least one hazard, otherwise ARAP warns.

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STPA

STPA — descent clearances (Manual §13.2)

Worked examples — all twelve methods · Manual chapter 13 · template v1.0

2 participants

Saved

1 · Purpose

2 · Control structure

3 · Unsafe control actions (5)

4 · Loss scenarios (2)

Traceability

Control action: procedures, rosters

Not providing causes hazard

Providing causes hazard

Too early / too late / wrong order

Stopped too soon / applied too long

Control action: Descent clearance

Not providing causes hazard

UCA-1

ATCO does not issue descent

when the aircraft must leave its level to avoid a conflict ahead

H-1 ×

remove

+ UCA

Providing causes hazard

UCA-2

ATCO clears descent through an occupied level

when unseparated traffic is at that level

H-1 ×

remove

UCA-3

ATCO clears descent below MVA

when vectoring over high terrain

H-2 ×

remove

+ UCA

Too early / too late / wrong order

UCA-5

Descent issued before preceding aircraft vacated level

when relying on the label trend

H-1 ×

remove

+ UCA

Stopped too soon / applied too long

+ UCA

Control action: Execute descent

Not providing causes hazard

Providing causes hazard

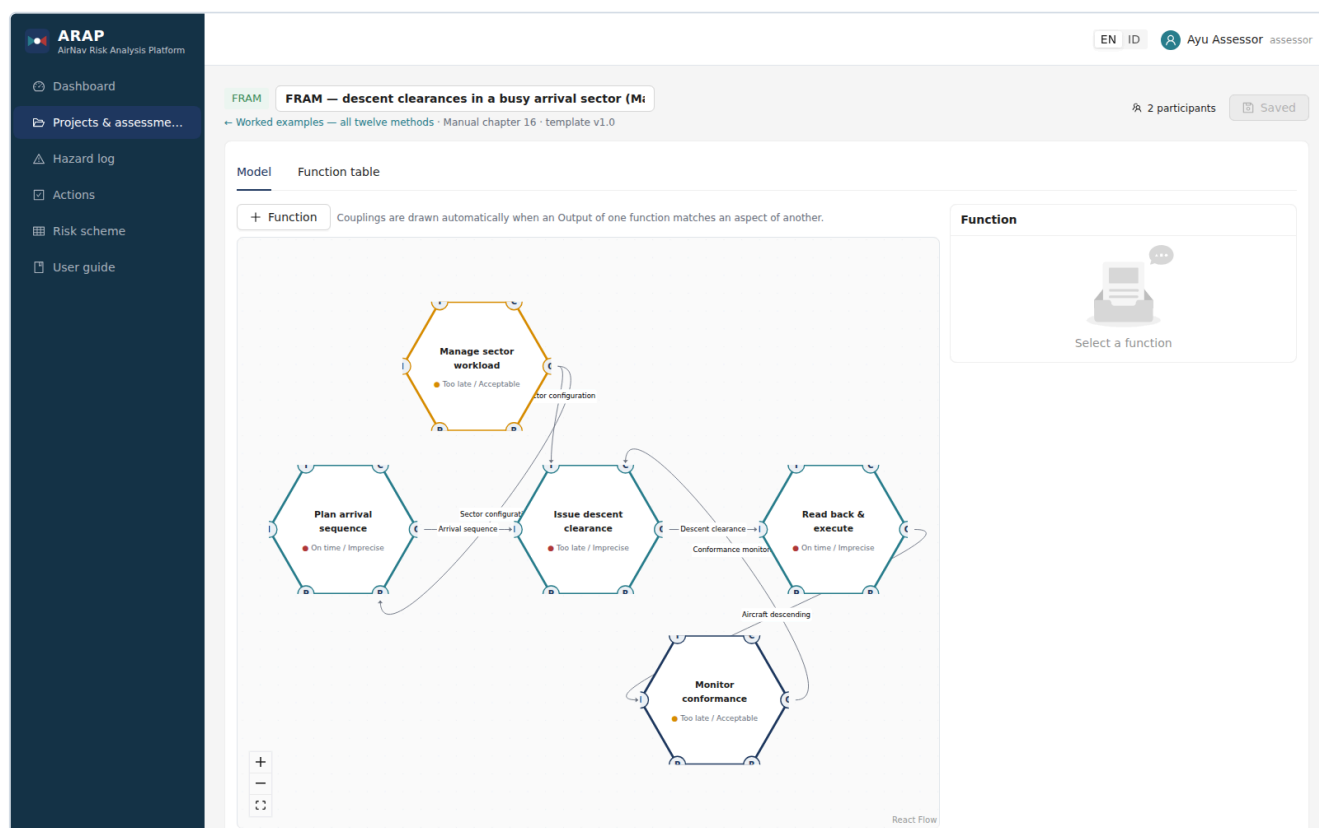
Too early / too late / wrong order

Stopped too soon / applied too long

UCA table

- Loss scenarios** — for each UCA, the scenario type, description and the resulting requirement or constraint. **Send hazards to hazard log** creates one hazard-log entry per STPA hazard.
- Traceability** — the chain loss → hazard → constraint/UCA → scenario → requirement, with broken chains flagged.

12. FRAM



FRAM model

- **+ Function** adds a function (hexagon). Drag to arrange.
- Select a function to set its name, type (human, technological, organisational) and the six **aspects** — Input, Output, Precondition, Resource, Control, Time. Type an entry and press Enter; existing outputs are offered as suggestions.
- **Couplings are drawn automatically** wherever an Output of one function has the same text as an aspect of another. Unmatched aspects are listed so you can fix typos or add missing functions.
- Record the **output variability** (timing: too early / on time / too late / not at all; precision: precise / acceptable / imprecise). The dot colour on each hexagon shows it.
- The **Function table** tab lists everything in one table.

13. Extended methods (Manual chapters 18–30)

Version 0.2 adds thirteen methods. They all work like the others: open the study, edit, press **Calculate** where there is one, and **Save**. The demo project **DEMO-02** has one worked example of each, matching the Manual.

13.1 Collision risk — CRM (Manual §18)

ARAP AirNav Risk Analysis Platform

EN ID Ayu Assessor assessor

Collision risk modelling CRM — RVSM vertical risk and parallel-route spacing

Worked examples — thirteen extended methods · Manual chapter 18 · template v1.0

2 participants Save

Reich collision risk model in occupancy form (ICAO Doc 9689 / Doc 9574). Overlap probabilities and occupancies should come from regional monitoring agency data (e.g. height-keeping and lateral deviation monitoring).

Vertical (RVSM) Lateral (route spacing)

Parameters

λx length (ft) 184 λy wingspan (ft) 175

λz height (ft) 50

Lateral deviation model Double expone... scale (NM) 0.9

Route spacing S_y (NM) 15

P_z(0) — vertical overlap, same level 0.52 E_y(same) — same-direction lateral occupancy 0.1

E_y(opp) — opposite-direction lateral occupancy 0 S_x — longitudinal window (NM) 80

|Δv| (kt) 20 |v| (kt) 470

|z| (kt) 1.5 |y(S_y)| lateral closing speed at loss of spacing (kt) 4

Target level of safety 0.000000005

Calculate

Result

P_y(S_y) at 15 NM	1.63e-8
Lateral risk (fatal accidents / fh)	1.578e-10 Meets TLS
Minimum spacing meeting TLS	11.68 NM

collision risk / fh

route spacing (NM)

min 11.7 NM

CRM lateral spacing

Two tabs. **Vertical (RVSM)** takes the Reich-model parameters: overlap probabilities, occupancies, speeds and aircraft dimensions (entered in feet). **Calculate** shows the same- and opposite-direction risk, the total and whether it meets the target level of safety (TLS). **Lateral (route spacing)** also asks for the lateral-deviation model (double exponential or Gaussian) and its scale. It plots risk against route spacing on a log scale and gives the **minimum spacing** that meets the TLS. The demo gives 1.87×10^{-9} (vertical) and 11.7 NM (lateral).

13.2 Event tree analysis — ETA (Manual §19)

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Ayu Assessor assessor

Event tree analysis

ETA — conflict after a level bust (Manual §19.3)

2 participants Save

Event tree analysis

ETA — conflict after a level bust (Manual §19.3)

Worked examples — thirteen extended methods · Manual chapter 19 · template v1.0

Initiating event

Conflict following a level bust

Frequency 0.025 per year

A successful barrier ends the sequence (barrier event tree)

Barriers / functional events (in order)

ID	Barrier	P(success)
B1	ATCO detects and resolves	0.90
B2	STCA alert and ATCO resolution	0.90
B3	ACAS II RA followed	0.90
+ Add		

Conditional probabilities (depend... path prefix e.g. "F" or "F-F")

After path	Barrier	P(success)
F	B2	0.80
+ Add		

Calculate

Initiating event

Conflict following a level bust

$f = 0.0250$ per year

B1

ATCO detects and resolve

B2

STCA alert and ATCO reso

B3

ACAS II RA followed

Up (green) = barrier succeeds · down (red) = barrier fails

Outcomes

Sequence	Outcome	Severity	Probability	Frequency (per year)
S	Resolved by ATCO — no loss of separation	D	0.900	0.0225
F-S	Loss of separation, resolved after STCA	C	0.0800	2.000e-3
F-F-S	Serious loss of separation, resolved by ACAS	B	0.0180	4.500e-4
F-F-F	Near mid-air collision / collision	A	2.000e-3	5.000e-5

Σ probability = 1.000000 · by severity: D 0.0225 · C 2.00e-3 · B 4.50e-4 · A 5.00e-5 · edit outcomes then recalculate

Event tree

1. Enter the initiating event and its frequency.
2. List the barriers **in the order they act**, with their probability of success.
3. Under **Conditional probabilities**, enter any dependence as a path prefix. For example, **F / B2 / 0.8** means "STCA succeeds 80% of the time when the controller has already failed".
4. Press **Calculate**. ARAP draws the tree and lists each sequence with its probability and frequency.
5. Name each outcome and give it a severity, then recalculate. The totals by severity appear under the table.

13.3 Human reliability — HRA (Manual §20)

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Human reliability (HEART / CARA)

HRA — controller and technician tasks (Manual §20.4)

2 participants

Save

HEP = $GTT \times \prod[(EPC - 1) \times APOA + 1]$. Use CARA for controller tasks and HEART for other tasks (e.g. engineering). Justify every APOA; use the HEP in FTA, LOPA or ETA.

Tasks

ID

Task

GTT

HEP

T1

ATCO acts correctly on an STCA alert

CARA B3

2.016e-3

T2

ATCO detects a level bust by monitoring the display (no alert)

CARA B1

0.0575

T3

Technician restores radar to operational configuration after maintenance

HEART F

0.0108

Calculate all (server)

Task T1

ATCO acts correctly on an STCA alert

CARA (ATC)

HEART

B3 — Respond to unique and trusted audible and visual i...

Error-producing conditions

EPC	Condition	Max	APOA	Factor
17	Traffic complexity	×10		2.80
6	Cognitive overload	×6		1.50
13	Low vigilance	×3		1.20

Add an error-producing condition

Nominal HEP 0.0004 → assessed HEP 2.016e-3

Justification of the task type and APOA values

HRA

Add a task and choose **CARA** (controller tasks) or **HEART** (other tasks) and the generic task type. Add error-producing conditions from the list. For each one, set the **APOA** slider (0–1) and write a justification. The HEP updates as you edit. **Calculate all** runs the server engine for the record.

13.4 Occurrence risk classification — ERC and RAT (Manual §21)

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Occurrence risk classification
Occurrence risk classification — ERC and RAT (Manual §
2 participants
Saved

Occurrences

ID

Occurrence

ERC

RAT

Class

OCC-101	Level bust FL240→FL23...	102	7	B
OCC-102	Vehicle entered runway...	50	—	
OCC-103	Frequency congestion ...	1	—	

Send red-band occurrences to hazard log

OCC-101

Level bust FL240→FL232 with crossing traffic; STCA; ATCO resolved; 2.8 NM / 700 ft

Select date

ESARR 2 severity class (analyst)

B

ARMS ERC

RAT scoring

Minimum separation achieved

Separation ≥ minimum (0)

> 75% of minimum (1)

> 50% and ≤ 75% (3)

> 25% and ≤ 50% (7)

≤ 25% of minimum (10)

Rate of closure

None (0)

Low (≤ 85 kt, ≤ 1000 ft/min) (1)

Medium (≤ 205 kt, ≤ 2000 ft/min) (2)

High (≤ 700 kt, ≤ 4000 ft/min) (4)

Very high (> 700 kt, > 4000 ft/min) (5)

Conflict detection (ATM ground)

Detected (0)

Detected late (1)

Not detected (2)

Not applicable (0)

Planning (ATM ground)

Plan correct (0)

Plan inadequate (1)

No plan (2)

Not applicable (0)

Execution

Execution correct (0)

Execution inadequate (1)

No execution (2)

Not applicable (0)

Ground safety net (STCA / MSAW / APW)

Triggered (0)

Not triggered (2)

Not applicable (0)

Recovery

Recovery correct (0)

Recovery inadequate (1)

No recovery / worsened (2)

Not applicable (0)

Airborne safety net / pilot initiative

TCAS/GPWS or see-and-avoid effective (0)

No RA / warning (2)

Not applicable (0)

Risk of collision 5 + controllability 2 = severity score 7

Assign the ESARR 2 severity class with the EUROCONTROL RAT look-up table (not embedded in ARAP) and record it above.

Notes / rationale

ERC

- **Add an occurrence.** Add a short description, date and occurrence ID.
- **ARMS ERC.** Click the matrix cell that answers Q1 (most credible accident outcome) and Q2 (effectiveness of the remaining barriers). The band and recommended response are shown.
- **RAT scoring.** Pick one answer per item. ARAP adds the points for risk of collision and controllability. Record the ESARR 2 severity class that you read from the EUROCONTROL RAT table.
- **Send red-band occurrences to hazard log.** Creates hazard entries for the red-band occurrences.

13.5 Safety argument — GSN (Manual §22)

ARAP
AirNav Risk Analysis Platform

EN ID Ayu Assessor assessor

Safety case (GSN) **Safety argument — ADS-B introduction (Manual §22.3)** 2 participants Saved

← Worked examples — thirteen extended methods · Manual chapter 22 · template v1.0

Structure valid 6 goals 4/4 solutions linked to evidence 1 undeveloped

Solid arrow = SupportedBy · open arrow, dashed = InContextOf · ◊ = undeveloped. Click an element to edit.

G1 Introduction of ADS-B surveillance in en-route sector II is acceptably safe

C1 Scope, environment and assumptions of assessment DEMO-G1

G2 Every FIA failure condition meets its safety objective

S1 Argue over the specification, the implementation and the transition

G6 Transition to operations does not increase controller workload beyond acceptable limits

S2 Argue over each failure condition FC-1 ... FC-3

G3 FC-3 (undetected erroneous position, one aircraft) $\leq 1 \times 10^{-4}$ per operating hour

G4 FC-4 (undetected erroneous position, several aircraft) $\leq 1 \times 10^{-4}$ per operating hour

Sn1 FIA worksheet — objectives (Risk #7 · open)

Sn2 FTA result for the integrity-monitoring architecture (Risk #9 · open)

Sn3 Hazard log entry HZ-0003 with acceptance... (Risk #003 · open)

J1 Acceptance authorities follow Manual Table 3.3

Sn4 Real-time simulation results (workload, R/T occupancy) (Risk #22 · open)

Sn4 — solution

Real-time simulation results (workload, R/T occupancy)

Parent G6

Evidence

ARAP study

#22 SIM — Simulation — new arrival procedu...

Status: **SIM #22 · open** · open study

Evidence is not yet complete / accepted — the argument is provisional.

Argument checks

✓ No structural issues found.

React Flow

GSN

- **Build the argument.** Click an element to edit it. Use **+ goal** / **+ strategy** / **+ solution** / **+ context** / **+ assumption** / **+ justification** to add elements below it. ARAP lays out the diagram automatically.
- **Link evidence.** For a **solution**, link the evidence: an ARAP study, a hazard log entry or an external document. The solution shows the evidence status in green (complete or accepted) or red (still provisional).
- **Argument checks.** This panel lists structural problems: more than one top goal, unsupported goals, strategies supported by something other than goals, solutions without evidence, and cycles. Mark a goal **undeveloped** when its support is still to come.

13.6 Common cause analysis — CCA (Manual §23)

ARAP AirNav Risk Analysis Platform

EN ID Ayu Assessor assessor

Common cause analysis Common cause analysis — ACC surveillance chain (Man) 2 participants Save

Worked examples — thirteen extended methods · Manual chapter 23 · template v1.0

Every AND gate or redundancy claim in an FTA, RBD or FHA rests on independence. CCA checks it three ways: where items are installed (ZSA), what single events can hit several at once (PRA) and what they share by design (CMA).

3 independence claim(s) not supported 0 claim(s) not yet assessed 3 particular risk(s) defeat redundancy Send failures of independence to hazard log

Zonal safety analysis (3) Particular risks (4) Common mode analysis (3)

+ Add row Delete selected Import claims from FTA

ID	Independence claim	Items claimed indepe...	Common-mode...	Analysis	Indepen...	Action
CM-01	☐ Main SDP (E5) and fallback (E6) fail independently	E5, E6	Software	Fallback runs the same software baseline as the main system	No	Diverse fallback or β -factor in FTA
CM-02	☐ Mains (E7), UPS (E8) and generator (E9) fail independently	E7, E8, E9	Location	UPS and generator switchgear share the power room (fire zone)	Partly	Include fire CCF in FTA
CM-03	☐ Radar A, radar B and ADS-B inputs are independent	E1, E2, E3	Shared utilities (power, HVAC, network)	All three use the single surveillance WAN (E4)	No	Diverse data path (FTA shows 82% of risk)

CCA

Three worksheets: **Zonal safety analysis**, **Particular risks** and **Common mode analysis**. On the common-mode tab, **Import claims from FTA** creates one independence claim for every AND or k-out-of-n gate in a chosen fault tree. For each claim, record the common-mode source and whether independence holds. The tags at the top count unsupported claims and particular risks that defeat redundancy. **Send failures of independence to hazard log** passes them on.

13.7 RBD and Markov (Manual §24)

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RBD and Markov analysis

RBD and Markov — sector VHF availability (Manual §24.

2 participants

Save

Reliability block diagram

Markov model

series: Sector frequency available at CWP

parallel: Radio chains

Main Tx/Rx chain

MTBF 20000 h · MTTR 4 h

A = 0.999800

Standby Tx/Rx chain

MTBF 20000 h · MTTR 4 h

A = 0.999800

parallel: Links to remote site

Leased line

MTBF 4000 h · MTTR 8 h

A = 0.998004

VSAT backup

MTBF 3000 h · MTTR 12 h

A = 0.996016

VCCS

MTBF 50000 h · MTTR 2 h

A = 0.999960

parallel: Power

Mains

MTBF 2000 h · MTTR 2 h

A = 0.999901

UPS

MTBF 30000 h · MTTR 8 h

A = 0.999733

Calculate availability

Blocks are assumed independent — check with common cause analysis.

Availability	0.99995174
Unavailability	4.826e-5
Expected downtime	0.423 h/yr (25.4 min)

VCCS

Leased line

VSAT backup

Mains

UPS

Main Tx/Rx chain

Standby Tx/Rx chain

downtime saved if perfect (min/yr)

0 5 10 15 20 25

Selected element

Label Sector frequency available at CWP

series parallel koon block

Add inside: + block + series + parallel + koon

RBD

- **RBD tab.** Click a group or block to edit it. Groups can be **series**, **parallel** or **k-out-of-n**. Blocks have MTBF and MTTR. Add elements inside a group with the + buttons.
- **Calculate availability.** Gives availability, unavailability and downtime per year, plus a chart of how many minutes a year each block would save if it were perfect.
- **Markov model tab.** Define states (tick *Service up?*) and transitions with rates per hour, then press **Solve**. You get steady-state probabilities, availability, failure frequency, system MTBF, mean down time and mean time to first failure.

ARAP

AirNav Risk Analysis Platform

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RBD and Markov analysis

RBD and Markov — sector VHF availability (Manual §24.)

Worked examples — thirteen extended methods · Manual chapter 24 · template v1.0

2 participants

Save

Reliability block diagram

Markov model

Use a Markov model where blocks are not independent: shared repair crews, cold/warm standby, imperfect switchover (coverage). Rates are per hour ($\lambda = 1/\text{MTBF}$, $\mu = 1/\text{MTTR}$).

States

ID	Description	Service up?
S2	Main in service, standby ready	☒
S1	On standby, main in repair	☒
S0	No radio chain in service	☐
+ Add		

Transitions

From	To	Rate (/h)	Meaning
S2	S1	0.000049504	main fails, switchover OK
S2	S0	0.000000504	main fails, switchover fails
S1	S0	0.00005	standby fails
S1	S2	0.25	main repaired
S0	S1	0.25	one chain restored
+ Add			

Initial state (for MTFF) Solve

Availability	0.999997960
Unavailability	2.0396e-6 (1.07 min/yr)
Failure frequency	5.099e-7 /h
System MTBF	1,961,176 h
Mean down time	4.00 h
MTTFF	1,961,565 h (223.9 yr)

State	Description	π (steady state)
S2	Main in service, standby ready	1.00
S1	On standby, main in repair	1.9996e-4
S0	No radio chain in service	2.0396e-6

Markov

13.8 SWIFT (Manual §25)

A worksheet like HAZID. Each row is a *what if...?* question under a prompt category, with consequence, safeguards, severity, likelihood and recommendation. **Prompt category coverage** shows which categories have been considered. Use it to record "considered — nothing credible" for the rest. Rows with a consequence can be sent to the hazard log.

13.9 Hierarchical task analysis — HTA (Manual §26)

The screenshot shows the ARAP (AirNav Risk Analysis Platform) interface. On the left is a dark sidebar with navigation links: Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme, and User guide. The top right shows user information: EN ID, Ayu Assessor, and a Saved button. The main content area is titled 'Hierarchical task analysis' and 'HTA — respond to an STCA alert (Manual §26.3)'. It indicates 2 participants and 14 tasks, with 6 error modes. A 'Linked HRA study' dropdown shows '#16 HRA — controller and technician tasks (M...)' with an 'open' button. Below this are 'Tree' and 'Outline' tabs. The 'Tree' tab is active, showing a hierarchical diagram of tasks. Task 0, 'Respond to an STCA alert', is the root. It has four main sub-tasks: 1. Detect the alert, 2. Assess the situation, 3. Issue avoiding action, and 4. Monitor and restore normal operations. Each of these has further sub-tasks. For example, Task 1 has sub-tasks 1.1 (See / hear the alert) and 1.2 (Identify the aircraft pair). Task 2 has 2.1 (Check levels, tracks and clearances) and 2.2 (Decide the resolution). Task 3 has 3.1 (Transmit instruction) and 3.2 (Obtain and check read-back). Task 4 has 4.1 (Monitor separation), 4.2 (Issue further clearances), and 4.3 (Report the occurrence). The tasks are color-coded: blue for main tasks and yellow for sub-tasks. A 'Task 0' panel on the right allows for adding sub-tasks, defining error modes (using SHERPA-style taxonomy), and linking the task to an HRA study. The bottom right corner of the diagram area says 'React Flow'.

HTA

- **Build the hierarchy.** Select a task and press **Sub-task** to add a step below it. Numbering is automatic (1, 1.1, 1.2 ...).
- **Add plans.** Write a **plan** for every task that has sub-tasks. Tasks without one are flagged in red.
- **Record error modes.** Add error modes to bottom-level tasks.
- **Link to HRA.** Choose a **linked HRA study** at the top, then link a task to one of its HRA tasks to show the HEP.
- **Change the view.** The **Outline** tab shows the same content as an indented table.

13.10 Simulation (Manual §27)

ARAP

AirNav Risk Analysis Platform

EN ID

Ayu Assessor assessor

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Simulation (RTS / FTS)

Simulation — new arrival procedure (Manual §27.4)

2 participants

Save

Worked examples — thirteen extended methods · Manual chapter 27 · template v1.0

Exercise plan

Title RTS of new arrival procedure (sector APP-N)

Real-time simulation

Safety objectives / hypotheses

Show the new procedure does not increase workload or R/T occupancy and does not add conflicts.

Scenarios, traffic samples, runs and design

Peak traffic 38 arrivals/h; 8 runs per condition; 4 ATCO teams; counterbalanced order

Participants (ATCOs, pseudo-pilots)

8 current APP ATCOs, 4 pseudo-pilots

Limitations and validity (realism, learning effects)

Measures and run data

one value per run, comma-separated

ID	Measure	Unit	Better	Criterion	Value	Baseline
M1	Workload (ISA, 1-5)	ISA	lower	threshold	3.5	3.4, 3
M2	STCA alerts per hour	/h	lower	no worse than baseline (+margin)	0	1.2, 0
M3	R/T occupancy	%	lower	threshold	60	58, 63

+ Add

Analyse

Results

All success criteria met.

Measure	Baseline mean ± sd (n)	Solution mean ± sd (n)	Solution 95% CI	Δ	Welch's t-test
M1 Workload (ISA, 1-5), executive ATCO	3.69 ± 0.20 (8)	3.16 ± 0.17 (8)	3.02 – 3.30	-0.52	<0.0
M2 STCA alerts per hour	1.10 ± 0.23 (8)	0.85 ± 0.16 (8)	0.72 – 0.98	-0.25	0.02
M3 R/T occupancy	61.63 ± 2.67 (8)	51.75 ± 2.12 (8)	49.98 – 53.52	-9.88	<0.0

% of baseline mean

Simulation

- **Plan the exercise.** Record the exercise plan: objectives, scenarios, participants and limitations.
- **Define the measures.** For each measure, set whether lower or higher is better and a **criterion**. Use *threshold* (e.g. ≤ 3.5), or *no worse than baseline*, which takes an optional margin. Set the criteria **before** entering results.
- **Enter the run results.** Enter them as comma-separated numbers for the baseline and the solution.
- **Analyse.** You get the mean, standard deviation, 95% confidence interval, Welch's t-test and whether each criterion is met. You can send failed criteria to the hazard log.

13.11 Expert judgement (Manual §28)

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Structured expert judgement

Expert judgement — ATCO detection probability (Manu...

2 participants

Saved

Classical model (Cooke)

Delphi

Cooke's classical model: each expert gives 5/50/95% quantiles for seed questions (true value known to the facilitator) and target questions. Weights = calibration x information; experts below the significance level α get zero weight.

Expert name

+ Expert

+ Question

α 0.00

4 seed · 1 target

Calculate weights

ID	Question	Seed ?	Realisatio n	Scale	Expert A			Expert B			Expert C		
					5%	50%	95%	5%	50%	95%	5%	50%	95%
S1	Level busts reported in sector X last year	☒	14	linear	8	12	18	5	10	30	20	25	35
S2	STCA alerts per 1000 flights (sector X)	☒	2.1	linear	1	2	3	0.5	1.5	5	3	4	6
S3	Median ATCO response time to STCA (s)	☒	9	linear	6	9	14	4	8	20	12	15	20
S4	Read-back errors per 1000 clearances	☒	3.5	linear	2	3.5	6	1	4	10	6	8	12
Q1	P(ATCO does not resolve a level-bust conflict before STCA)	☐	target	log	0.03	0.08	0.2	0.01	0.1	0.4	0.1	0.2	0.4

Expert scores

Expert	Calibration	Information	Bins	Weight
Expert A	0.839	0.771	0/2/2/0	0.814
Expert B	0.596	0.249	0/1/3/0	0.186
Expert C	2.54e-5	0.562	4/0/0/0	0.000

Decision maker (combined distribution)

Item	Performance-weighted 5 / 50 / 95%	Equal-weight 5 / 50 / 95%
S1 seed Level busts reported in sector X last year	5.86 / 11.9 / 26.1	5.66 / 16.3 / 33.6
S2 seed STCA alerts per 1000 flights (sector X)	0.658 / 1.97 / 4.35	0.626 / 2.68 / 5.73

Expert judgement

- Classical model (Cooke).** 1. Add experts and questions. 2. Tick **Seed?** for calibration questions and enter the true value. 3. Enter each expert's 5%, 50% and 95% values. Quantiles that are incomplete or not increasing are flagged. 4. Press **Calculate weights** to get calibration, information and weight per expert, and the combined (decision-maker) distribution for every question.
- Delphi.** Add panellists and rounds and enter the estimates, then press **Summarise rounds** to see the median, quartiles and convergence per round.

13.12 Security risk (Manual §29)

ARAP

AirNav Risk Analysis Platform

EN ID

Ayu Assessor assessor

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Security risk assessment

Security risk assessment — surveillance and FDP (Mani

2 participants

Saved

Worked examples — thirteen extended methods · Manual chapter 29 · template v1.0

+ Add row

Delete selected

Send all rows to hazard log

Double-click a cell to edit. Tick rows to select.

I.		Asset	Threat scenario	Threat source	Vulnerability	C	I	A	L...	Security risk	Existing controls
SR-01	☐	ADS-B ground station / SDP input	Injection of false ADS-B targets (spoofing)	External attacker	ADS-B messages unauthenticated	1	5	3	3	15 Very high	Radar cross-check available; multilater
SR-02	☐	Flight data processing (FDP)	Ransomware encrypts FDP servers	External attacker	Flat network between office and ops systems	3	4	5	2	10 High	Backups; antivirus
SR-03	☐	VCCS maintenance access	Compromise of vendor remote-maintenance account	Supplier / third party	Shared vendor credentials	3	4	5	2	10 High	VPN
SR-04	☐	ATM adaptation data	Unauthorised change to sector or safety-net parameters	Insider (accidental)	No two-person rule on adaptation changes	1	5	2	2	10 High	Change log

Security risk = $L \times \max(C, I, A)$. High / very high: SR-01, SR-02, SR-03, SR-04. 4 threat scenario(s) have a safety effect — send those to the safety hazard log (ED-205 / Manual §29.3).

Security risk

A worksheet of threat scenarios: asset, threat, source, vulnerability, C/I/A impact (1–5), likelihood (1–5), controls, treatment and owner. **Security risk** = likelihood × the highest of C, I and A, shown as Low / Medium / High / Very high. Give each scenario a **safety effect** (severity A–E) where it has one. Only those rows are sent to the safety hazard log, so exploit details stay in the security study.

13.13 Occurrence investigation — SOAM, HFACS, Tripod Beta (Manual §30)

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Occurrence investigation Investigation OCC-102 — works vehicle on runway strip

2 participants Saved

Worked examples — thirteen extended methods · Manual chapter 30 · template v1.0

Occurrence SOAM HFACS Tripod Beta

Organisational factors

- Training (TR)**
Contractor drivers not trained on runway closure procedures
- Change management (CM)**
Works plan changed the closure time without a risk assessment
- Communication (CO)**
Revised closure time not briefed to contractor

Contextual conditions

- Workplace conditions**
Night works; poor signage visibility near K3
- Human performance limitations**
Driver's mental model: runway closed for works
- Physiological and emotional factors**
Driver at end of 12-hour shift

Human involvement

- Driver crossed the lit stop bar believing the runway was closed
- TWR did not challenge the driver's position report

Absent / failed barriers

- Awareness** - failed
Driver RT training and licence
- Restriction** - absent
Follow-me / escort for works vehicles
- Restriction** - failed
Stop bar at holding point K3

Occurrence

OCC-102
At night a works vehicle entered the runway strip of 07L without clearance. RIMCAS alerted; the TWR ATCO instructed an aircraft on 4 NM final to go around. Closes

Barriers that worked

- RIMCAS alert
- ATCO go-around instruction

Barriers link failed barriers to the bowtie they belong to

Barrier	Type	Status	Bowtie barrier
Driver RT training and licence	Awareness	failed	PB3 Driver RT training & licence — Runway incursion
Follow-me / escort for works vehicles	Restriction	absent	PB4 Follow-me / escort — Runway incursion
Stop			

Contextual conditions

Category	Condition
Workplace conditions	Night works; poor signage visibility near K3
Human performance limitations	Driver's mental model: runway closed for works
Physiological and emotional factors	Driver at end of 12-hour shift
+ Add	

Organisational factors

Factor	Finding
Contractor drivers not	

SOAM

- **Occurrence** — the facts, the counts of failed barriers, HFACS categories and actions, and **Send occurrence to hazard log**.
- **SOAM** — barriers (type and status), human involvement, contextual conditions, organisational factors and safety actions. The SOAM chart updates as you type. Link each failed or absent barrier to the **bowtie barrier** it corresponds to. The bowtie then shows "recorded as failed in N investigation(s)" when you select that barrier.
- **HFACS** — tick categories in the four tiers and record the evidence for each.
- **Tripod Beta** — event trios (agent, object, event) with failed or missing barriers traced to immediate cause, precondition, underlying cause and Basic Risk Factor. A BRF profile summarises them.

14. The hazard log

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+ New hazard

↓ CSV

Hazard log

Single register of hazards from every method study, with current and residual risk, controls and owners (Manual §18.2).

Search

Q

Risk region

6 of 6

Ref	Hazard	Unit	System	Owner	Initial	Residual	Controls	Review	Status
HZ-0001	Unauthorised entry onto active runway	Soekarno-Hatta TWR	Aerodrome	TWR unit chief	3A	2A	2	2026-10-18	open
HZ-0002	Level bust leading to potential mid-air collision	Jakarta ACC	ATM	ACC unit chief	3A	1A	2	2027-01-26	open
HZ-0003	Total loss of surveillance picture at ACC sector	Jakarta ACC	Surveillance	ACC unit chief	3B	3B	1	2026-11-12	open
HZ-0004	ATCO fatigue at end of night shift	Jakarta ACC	Roster	ACC unit chief	4C	3C	1	2026-09-23	open
HZ-0005	New site outside radio line of sight of low-level sector	Jakarta APP	VHF	ACC unit chief	3B	—	0	2026-11-27	open
HZ-0006	Loss of surveillance display at cut-over	Jakarta APP	Surveillance	Project manager	3C	2C	1	2026-12-27	open

< 1 >

Hazard log

The hazard log is the single register of hazards from all studies (Manual §31.2). Search, filter by risk region, sort, and export to **CSV**. Overdue review dates are shown in red.

Click a hazard to open it:

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Hazard log

Single register of hazards from every method study, with current

Search

Risk region

Ref	Hazard	Unit
HZ-0001	Unauthorised entry onto active runway	Soeka
HZ-0002	Level bust leading to potential mid-air collision	Jakarta
HZ-0003	Total loss of surveillance picture at ACC sector	Jakarta
HZ-0004	ATCO fatigue at end of night shift	Jakarta
HZ-0005	New site outside radio line of sight of low-level sector	Jakarta
HZ-0006	Loss of surveillance display at cut-over	Jakarta

×

HZ-0001 — Unauthorised entry onto active runway

Save

Details and risk

Controls (2)

Links

History

* Hazard

Unauthorised entry onto active runway

Causes / threats

Consequences

Unit

System

Owner

Soekarno-Hatta TWR

Aerodrome

TWR unit chief

Status

Review date

open

2026-10-18

Initial / current risk

Residual risk

clear

	Catastrophic Hazardous Major Minor Negligible				
	A	B	C	D	E
Frequent 5	5A	5B	5C	5D	5E
Occasional 4	4A	4B	4C	4D	4E
Remote 3	3A	3B	3C	3D	3E
Improbable 2	2A	2B	2C	2D	2E
Extremely improbable 1	1A	1B	1C	1D	1E

	Catastrophic Hazardous Major Minor Negligible				
	A	B	C	D	E
Frequent 5	5A	5B	5C	5D	5E
Occasional 4	4A	4B	4C	4D	4E
Remote 3	3A	3B	3C	3D	3E
Improbable 2	2A	2B	2C	2D	2E
Extremely improbable 1	1A	1B	1C	1D	1E

Rationale and evidence for the ratings (required for residual risk)

Occurrence data 2023-2025; barrier audit results

Description / context

Hazard detail

- **Details and risk** — description, causes, consequences, unit, system, owner, status and review date. Rate **initial/current** and **residual** risk by clicking a cell in each matrix. A residual rating requires a **rationale** (the basis of the judgement).
- When you create a new hazard, ARAP checks for **possible duplicates** as you leave the title field.
- **Controls** — each with side (prevention/recovery), type, status, effectiveness, owner and a safety-critical flag. Only *existing-verified* controls may be credited in the current risk.
- **Links** — the source study, the assessment and the related actions.
- **History** — every change from the audit trail.

15. Actions

ARAP
AirNav Risk Analysis Platform

EN ID Ayu Assessor assessor

Actions
Risk-reduction actions with owners and due dates. Overdue actions are escalated on the dashboard (COM-12).

+ New action

Ref	Action	Hazard	Owner	Due	Status
ACT-0003	Collect KSS survey for two roster cycles	HZ-0004	FRMS specialist	2026-09-18	open
ACT-0004	Audit stop-bar U/S procedure compliance	HZ-0001	Aerodrome ops	2026-09-25	In progress
ACT-0005	Radio coverage study for new site	HZ-0005	CNS engineering	2026-10-28	open
ACT-0001	Install diverse, physically separate surveillance data path	HZ-0003	CNS engineering	2026-12-27	open
ACT-0002	Implement SFL mismatch alert in ATM system	HZ-0002	ATM systems	2027-02-25	open

< 1 >

Actions

Actions have an owner, a due date, a status (open, in progress, closed) and an optional link to a hazard. Overdue actions are flagged here and on the dashboard. **Closing an action requires closure evidence** (e.g. a report number). A Viewer can update an action when the action's owner field is their username.

16. Review, approval and locking

Status	Meaning	Next step and who does it
Draft	Being prepared; editable.	Submit for review — assessor
In review	Reviewer checking.	Endorse or Return to assessor (comment required) — reviewer
Endorsed	Reviewed; locked.	Accept residual risk — authority for the worst residual region; or Reject (comment required)
Accepted	Residual risk accepted; locked.	Close — reviewer or authority
Rejected	Not accepted.	Revise and Submit again
Closed / Superseded	Finished, or replaced by a newer version.	—

Rules ARAP enforces

- An assessment with an **intolerable** residual risk cannot be accepted.
- An assessment with **no hazards** cannot be accepted.
- The person accepting must hold authority for the **worst** residual risk region in the assessment (e.g. *Tolerable (upper)* needs the Accountable Executive).
- Endorsed, accepted and closed assessments are **locked**: studies and hazards become read-only. To change anything, use **New version**. This copies the studies into a new draft (version n+1) and marks the old version *superseded*.

The screenshot shows the ARAP interface for a 'Transition safety assessment' (v1) in 'accepted' status. The assessment is titled 'Transition safety assessment' and is version 1. The status bar indicates 'Assessment accepted'. The review history shows an endorsement by a reviewer and an acceptance by the director.

Scope, environment and assumptions

Scope: Cut-over of APP positions, VCCS and surveillance feeds.
 Environment: —
 Assumptions: Cut-over at night, low traffic.
 Risk scheme: version 1

Method studies (1)

Method	Study	Status	Updated
HAZID	Transition HAZID	open	2026-09-28 10:22

Hazards (1)

Ref	Hazard	Initial	Residual	Controls
HZ-0006	Loss of surveillance display at cut-over	3C	2C	1

Risk acceptance

Worst current/residual region: **Tolerable (lower)**
 Acceptance authority: **Director of Operations / Engineering**

Actions (0)

Review and approval history

- endorse by reviewer
2026-09-28 10:23 · in_review → endorsed
- accept by director
2026-09-28 10:23 · endorsed → accepted
"Accepted subject to cut-over at night."

Accepted assessment

Every transition is recorded in the **review and approval history** and the audit trail.

17. Reports and exports

What	Where	Format
Safety Assessment Report (Manual §31.3)	Assessment page → Report (.docx)	Word
Hazard log	Hazard log → CSV	CSV (opens in Excel)
Bowtie diagram	Bowtie editor → PNG / SVG	Image

The report includes scope, methods used, hazards and their risk ratings, study results (fault tree cut sets, LOPA verdict, fatigue indicators, Bayesian network probabilities, and for the extended methods the event-tree outcomes, collision risk, HEPs, availability, simulation statistics, expert weights, occurrence classifications, the GSN argument and the investigation findings), actions, the approval history, and the versions of the risk scheme and calculation engines used.

18. Administration

18.1 Risk scheme

Risk scheme shows the matrix, the tolerability regions with their required action and acceptance authority, and the likelihood levels with their quantitative bands.

The screenshot displays the ARAP (AirNav Risk Analysis Platform) interface. The left sidebar contains navigation links: Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme (selected), and User guide. The main content area is titled 'Risk scheme' and includes a subtitle: 'Active version 1. Changes create a new version and never alter locked assessments (ADM-01).'.

Risk matrix

The risk matrix is a 5x5 grid. The columns are labeled 'Catastrophic', 'Major', 'Minor', and 'Negligible'. The rows are labeled 'Frequent', 'Occasional', 'Remote', 'Improbable', and 'Extremely improbable'. The cells are color-coded: red for 'Intolerable', orange for 'Tolerable (upper)', yellow for 'Tolerable (lower)', and green for 'Acceptable'. A callout box points to cell 5B, which is labeled '5B - Intolerable'.

Tolerability and acceptance authority

Region	Cells	Required action	Authority
Intolerable	3A, 4A, 4B, 5A, 5B, 5C	Do not introduce or continue; redesign or add controls.	Cannot be accepted
Tolerable (upper)	2A, 3B, 4C, 5D	Additional controls; ALARP justification; monitoring plan.	Accountable Executive
Tolerable (lower)	1A, 2B, 2C, 3C, 3D, 4D, 4E, 5E	Mitigate where reasonably practicable; document ALARP.	Director of Operations / Engineering
Acceptable	1B, 1C, 1D, 1E, 2D, 2E, 3E	Accept; monitor.	Unit / branch manager

Likelihood levels

Level	Name	Nama	Band (per operating hour)
5	Frequent	Sering	1e-3 and above
4	Occasional	Kadang-kadang	1e-5 to 1e-3
3	Remote	Jarang	1e-7 to 1e-5
2	Improbable	Tidak mungkin	1e-9 to 1e-7
1	Extremely improbable	Sangat tidak mungkin	< 1e-9

Risk scheme

Administrators can reassign cells: choose a region under **Paint region**, click the cells, then **Save as new version**. ARAP checks that every cell belongs to exactly one region. Existing assessments keep the version they were created with.

18.2 Users

Users (administrators only) lists accounts. Click one to edit, or **New user** to add one. Set the role, unit and — for authorities — the risk regions they may accept. Deactivate rather than delete leavers.

18.3 Audit trail

Audit trail (reviewers and administrators) lists every create, change, approval and export with the user, time and changed fields. Filter by entity.

19. Troubleshooting and FAQ

I can't edit a study or hazard. Either your role is Viewer or Authority, or the assessment is endorsed/accepted/closed. Use **New version** on the assessment page (assessor or reviewer).

"A rationale is required for every risk judgement". Fill in *Rationale and evidence* in the hazard before saving a residual rating.

"Residual risk region ... requires ...". You do not hold authority for the worst residual risk in the assessment. Ask the named authority to accept it.

The worksheet did not keep my edit. Finish the edit (Enter or click another cell), then press **Save** in the header.

My session ended. Sessions last 8 hours by default; sign in again.

Numbers differ slightly from the Manual. The engines reproduce the Manual's worked examples within the tolerances in SRS §9. Rounding in the Manual text is to 2–3 significant figures.

20. What version 0.2 does not do yet

Version 0.2 covers the core workflow and all twenty-five methods. The following SRS items are planned for later releases:

- Single sign-on through Keycloak / SAML / OIDC with MFA. Version 0.2 uses local accounts with JWT tokens.
- Real-time co-editing of a diagram by several users, and the offline workshop pack.
- Imports and exports in BowTieXP XML, Open-PSA MEF, XMLBIF, GeNIe and FRAM Model Visualiser formats.
- File attachments (MinIO), e-mail notifications and reminders.
- Integrations with the occurrence reporting, rostering and maintenance systems.
- CAST (STPA for accident analysis), noisy-OR helpers for Bayesian networks, Monte Carlo uncertainty, and the full “braked” sleep-recovery form of the fatigue model.
- An Apache Superset / Metabase reporting database.
- For the extended methods: import of occurrences from the reporting system into the ERC/RAT module, CRM sensitivity cases, administrator editing of the CARA/HEART libraries, and restricted (need-to-know) access to security studies (SRS SRA-04).